

wink n win

Compass



VCG 2026 Think-a-Thon

A major monetization opportunity already sits within Spotify's Free base

Spotify already has the audience, and converting even a small share of its Free base can create meaningful value.

SPOTIFY USER BASE

761M total monthly active users

61.5%

Ad-Supported · 468M listeners

38.5%

Premium · 293M subscribers

WHY CONVERSION MATTERS



Large Free base
Opportunity already exists



Premium economics
Higher gross margin



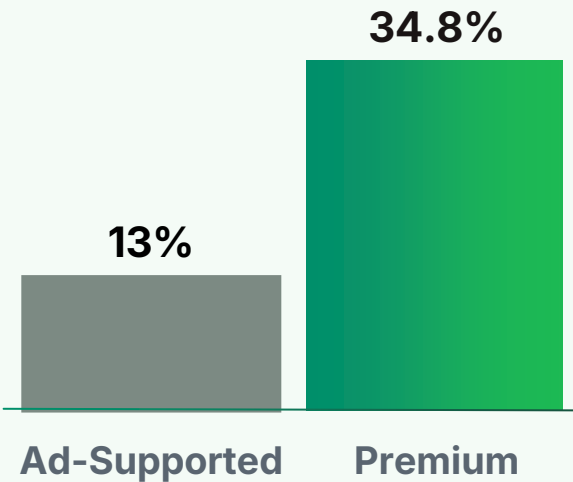
Monetization potential
More value from existing users

GROSS MARGIN ADVANTAGE

Premium delivers

~2.7x

gross-margin advantage versus Ad-Supported.

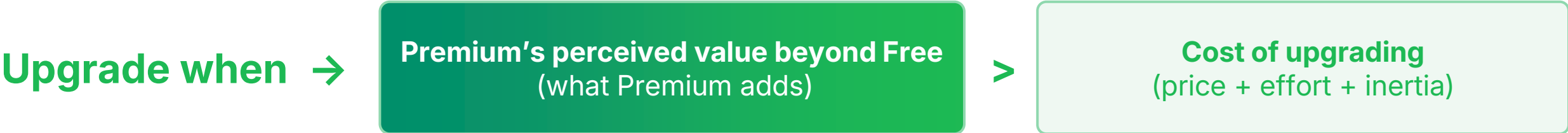


More Premium conversion strengthens Spotify's overall profitability.

KEY TAKEAWAY: Spotify's large Free base and stronger Premium economics create meaningful conversion headroom.

Spotify can influence the upgrade decision beyond broad changes to Free, price, or features

Users upgrade when Premium feels valuable enough beyond Free to justify the cost of upgrading.



BROAD CHANGES ACROSS USERS

Change product value or the cost of upgrading broadly.

-  **Degrade the Free-tier experience**
More upgrade pressure, but risks Free engagement and ad revenue.
-  **Add Premium features**
More potential value, but costs more without guaranteed conversion.
-  **Broad trials or discounts**
Lower barrier, but risk waste and promotion fatigue.

Can help, but broad use may create trade-offs and inefficiency when users respond differently.

USER-LEVEL FACTORS

Where Spotify can influence upgrade decisions more precisely.

-  **PERSUADABILITY** Whose upgrade decision can we change?
-  **TIMING** When does Premium matter most?
-  **RELEVANCE** Which Premium benefit matters most?
-  **SALIENCE** How do we make that value stand out enough to act on?



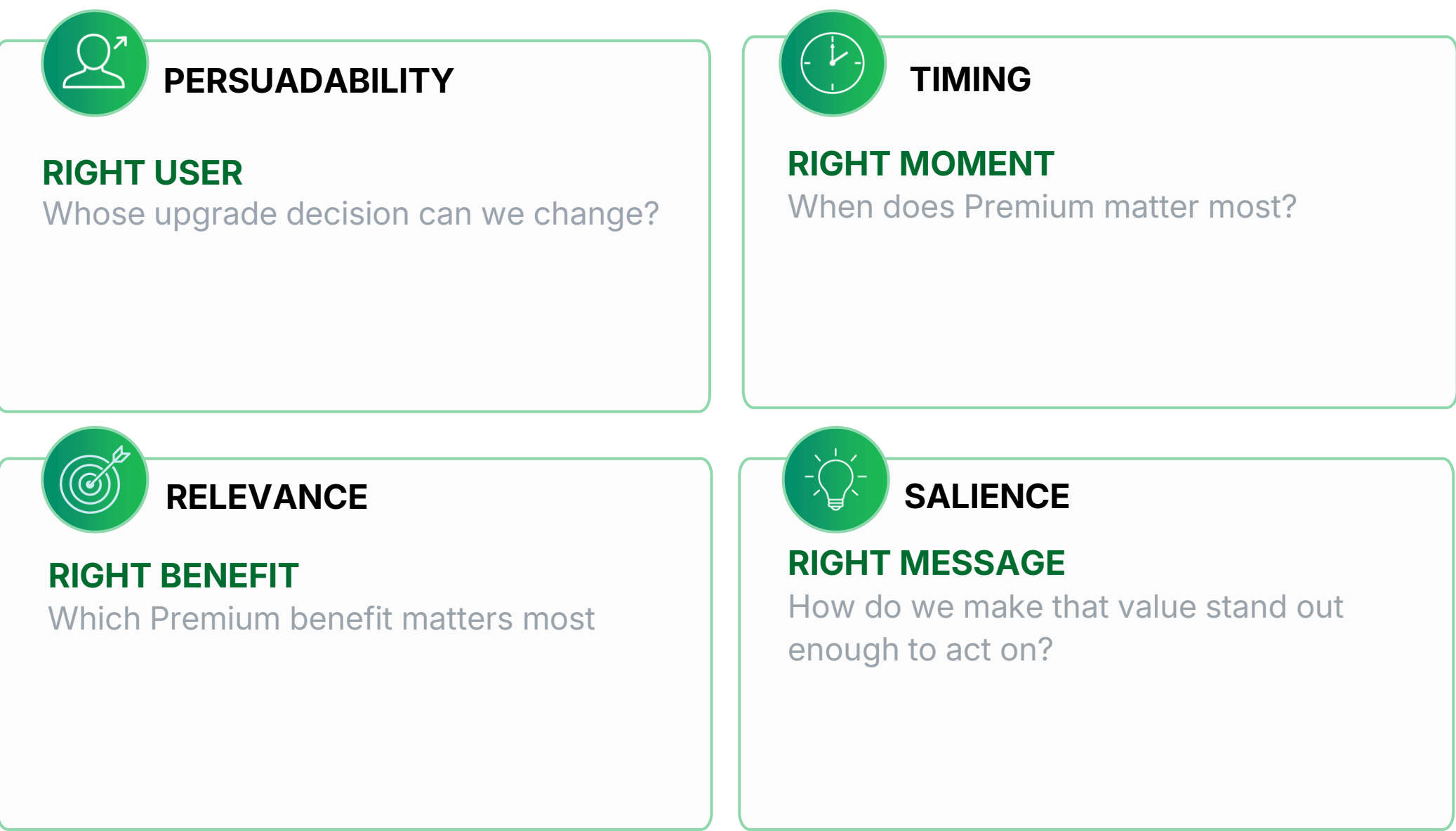
KEY TAKEAWAY:

Upgrade decision can be influenced more precisely through persuadability, timing, relevance, and salience.

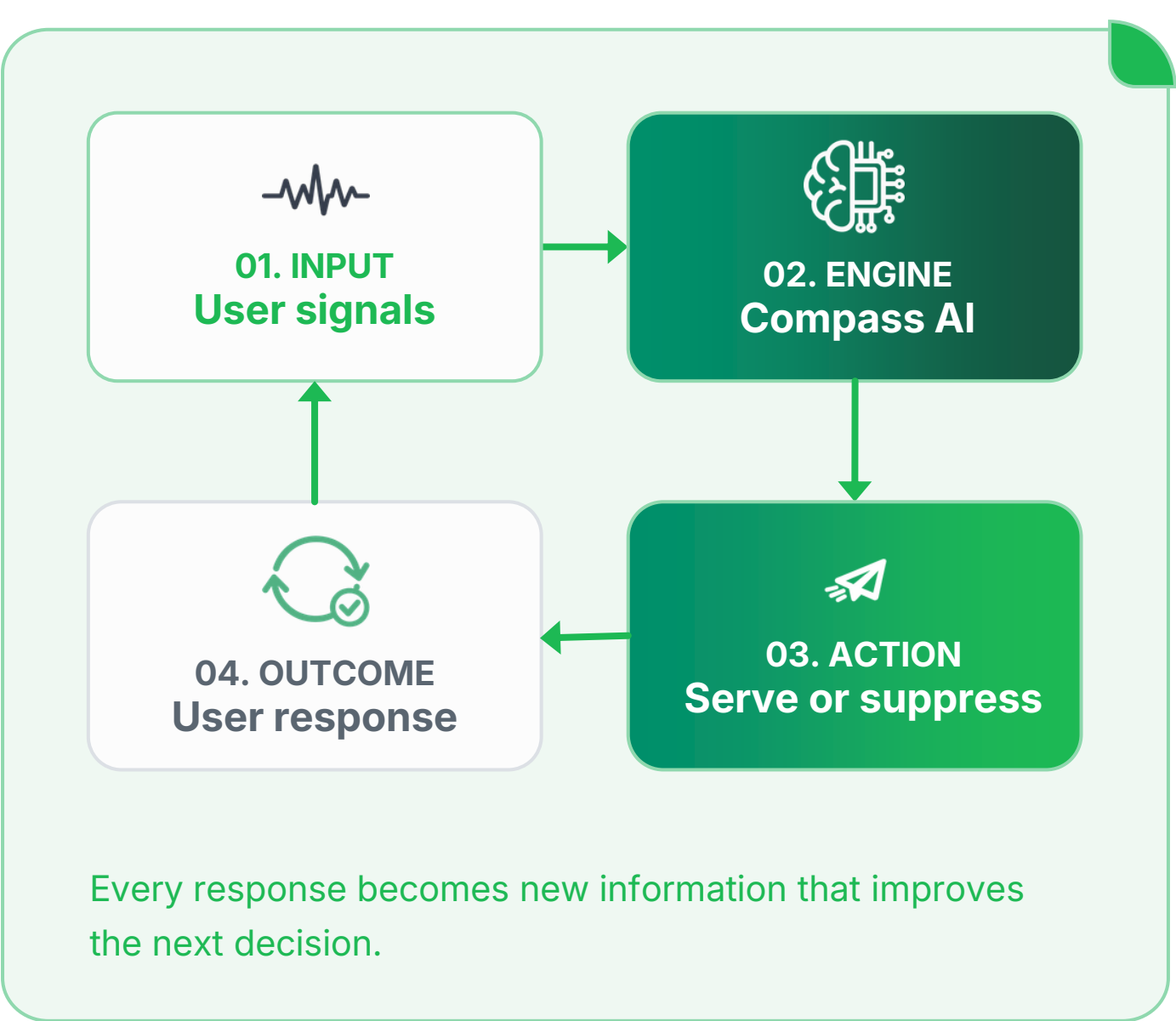
Compass AI improves Premium conversion through smart interventions

A behavioral decision layer decides who to approach, when to act, what to highlight, and how to communicate it.

FOUR DECISION FACTORS

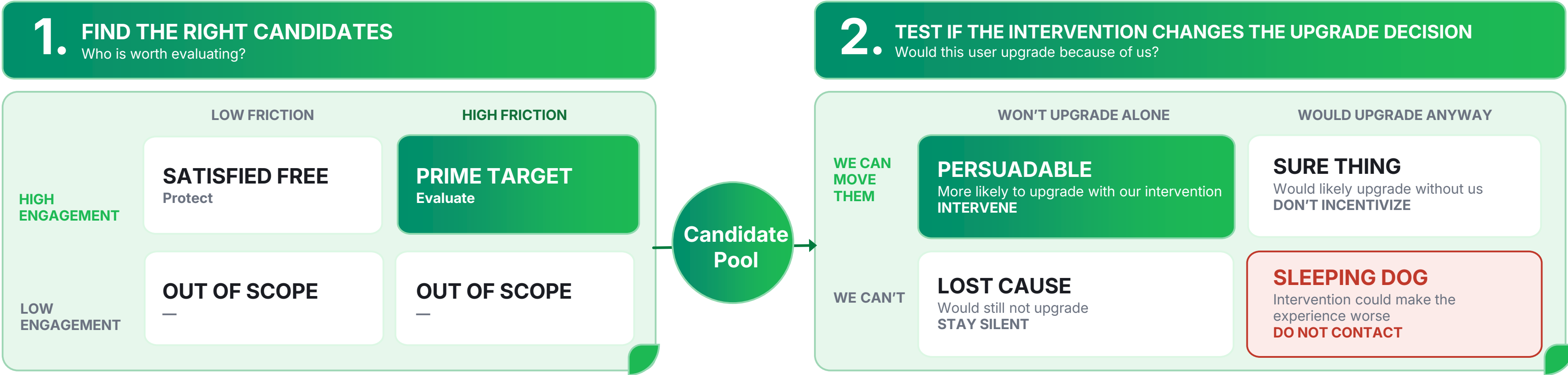


THE DECISION LOOP



Intervene only when Spotify can change the upgrade decision

A two-step filter finds promising candidates, then determines whose decision can actually change.



MEET MINH VY - A PERSUADABLE USER

MINH VY

High engagement + repeated skip-limit friction

→

WHY SHE QUALIFIES

Prime Target → Upgrade decision can change → Persuadable

→

ACTION

Show the Premium benefit matching her active need

Unlimited skips + Standard Premium

PERSUADABLE → INTERVENE

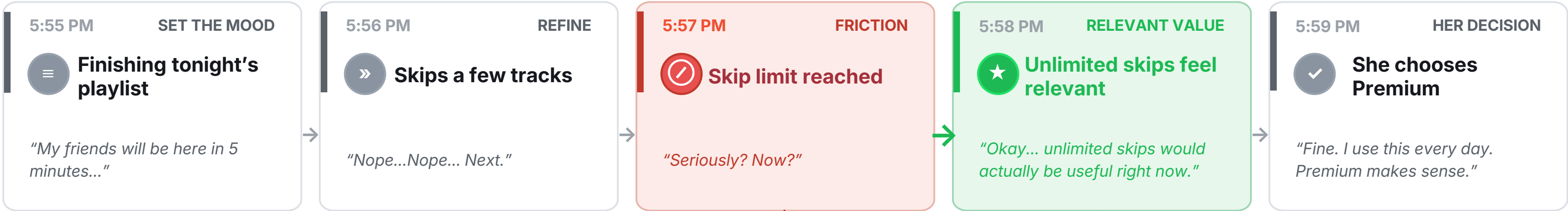
A real moment of friction can make the right Premium benefit immediately relevant

For Minh Vy, Compass AI connects playlist-control friction to the Premium benefit that addresses it.

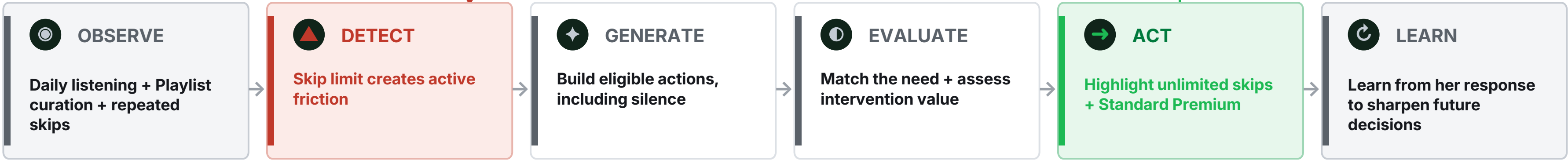
MINH VY, 20 **FREE USER · DAILY LISTENER · PLAYLIST CURATOR**

Friday 5:55 PM · Friends arrive in ~5 minutes · She is finishing her "Friday Dinner" playlist

WHAT MINH VY EXPERIENCES



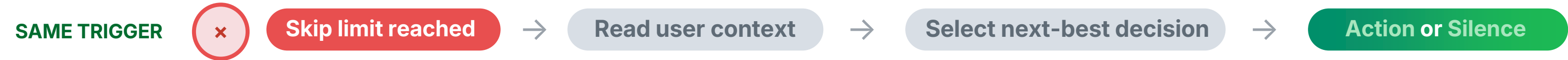
HOW COMPASS AI REACHES THE DECISION



KEY TAKEAWAY: Compass AI turns context into a specific next-best action and learns from the user's response.

Same trigger. Different users. Different next-best decisions

User context helps distinguish the underlying need behind the same trigger.



MINH VY
PLAYLIST CURATOR

Curating a playlist
• repeated skips

INFERRED NEED
Control is the active need

NEXT-BEST DECISION
PLAYBACK CONTROL

Unlimited skips
Standard Premium

THUY LINH
MUSIC EXPLORER

Discovery-heavy session
• repeated skips

INFERRED NEED
Discovery is the active need

NEXT-BEST DECISION
AI DJ DISCOVERY

Personalized discovery
1-month trial

TUAN NGHIA
PROMOTION FATIGUE

Recent offer dismissals
• repeated skips

INFERRED NEED
Avoid promotion pressure

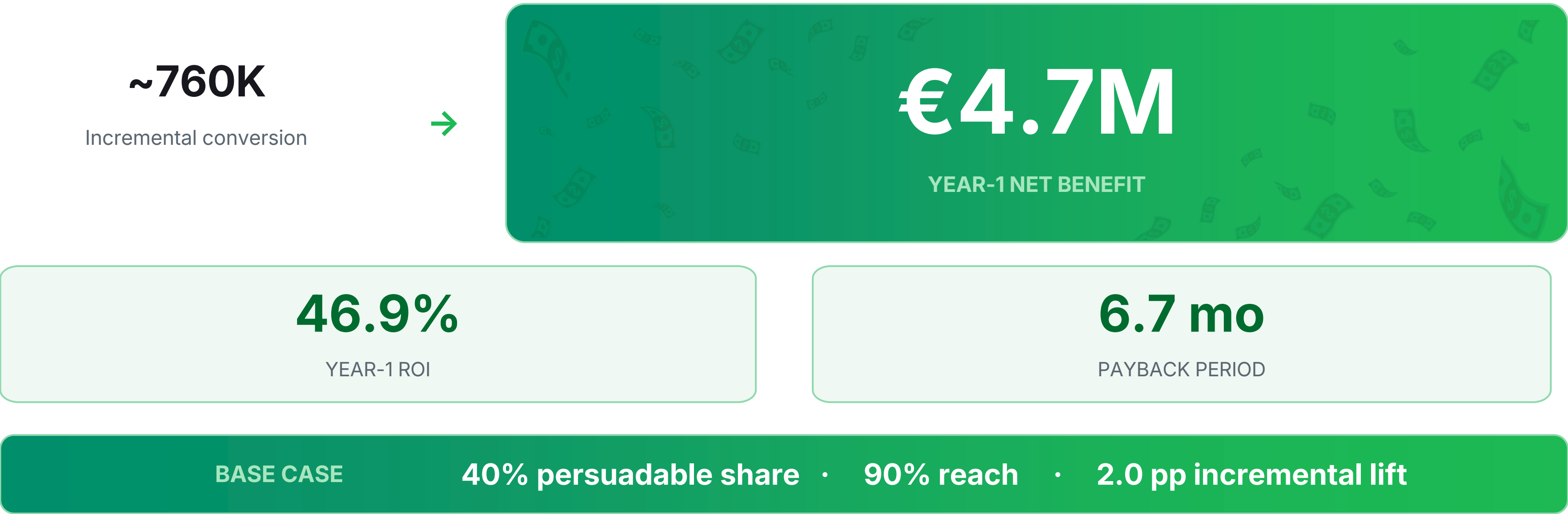
NEXT-BEST DECISION
DELIBERATE SILENCE

No promotion
Protect engagement

KEY TAKEAWAY: Personalization means choosing the right next-best decision, including staying silent.

Under the base case, Compass AI presents an attractive investment case.

~760K incremental conversion drive €4.7M Year-1 Net Benefit, with fast capital recovery.



Directional base case; full calculation and sensitivities are in the appendix.

 **KEY TAKEAWAY:** The base case shows attractive Year-1 returns, making Compass AI worth investing in.

Compass AI's returns depend on key assumptions that need testing

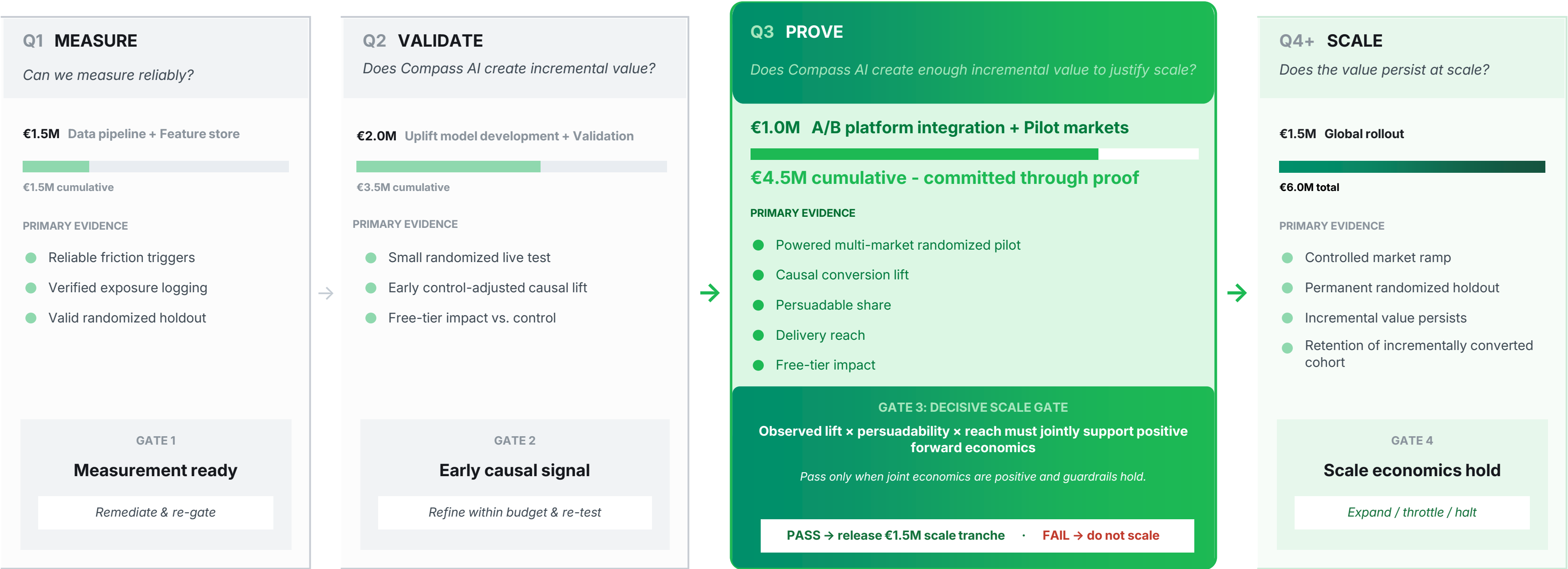
Each major risk has a clear mitigation, measurable guardrail, and predefined action.

RISK	MITIGATION	GUARDRAIL	ACTION
1. Incrementality is overstated	Persistent randomized control	Measured lift vs. holdout ≥ 1.33 pp at base share and reach	DO NOT SCALE
2. Eligible volume / delivery underperforms	Recalibrate targeting; improve eligible surfaces	$\geq 33\%$ persuadable $\geq 80\%$ reach	RE-TUNE / FIX DELIVERY
3. Free-tier is damaged	Caps, suppression, deliberate silence	Listening & ad revenue $\leq 1\%$ below control	THROTTLE
4. Incremental converts churn faster	Separate retention tracking and onboarding	Nudged-cohort churn $\leq 2.0\%$ / month	TIGHTEN OFFER / TARGETING
5. Privacy / trust / policy risk	Approved signals only; sensitive attributes excluded; privacy/legal review	Policy and eligibility checks clear before action	BLOCK / REDESIGN

 **KEY TAKEAWAY:** Measurable risks and predefined actions help Spotify identify issues early and respond before committing more investment.

Spotify can validate Compass AI step by step before committing to scale

Each phase provides the evidence needed to justify the next step.



Invest €4.5M in Compass AI now; scale only if results justify it.

Fund Q1-Q3 first; release the final €1.5M only if results support further investment.



WHY COMPASS AI ?

Built on Spotify's existing Premium value and behavioral context, Compass AI:

PERSONALIZE THE UPGRADE JOURNEY

Better selection · timing · benefit matching · deliberate silence

PROTECT SUSTAINABLE FREE GROWTH

Grow Premium without weakening the Free experience

1 APPROVE TODAY

€4.5M

Staged build envelope

€4.5M authorized through Q3

released in tranches to measure, validate and prove the business case

Q1

€1.5M

Q2

€2.0M

Q3

€1.0M

2 Q3 SCALE GATE

PROVE THE BUSINESS CASE

- Joint incremental economics vs. control clear the scale hurdle.
- Persuadable share $\geq 33\%$
- Delivery reach $\geq 80\%$
- Free-tier guardrails hold

PASS → SCALE

FAIL → DO NOT SCALE

3 UNLOCK AFTER PROOF

€1.5M

Final rollout tranche

+ ~€4.0M / year

Base-case operating cost at scale

Includes inference, model operations and offer subsidy.

€6.0M total build if Q3 passes



Thank You!



Appendix

Supporting analysis behind the business case

A1	Decision walkthrough	A5	Unit economics
A2	System architecture	A6	Cost & build recovery
A3	Conversion build	A7	Break-even & joint sensitivity
A3a	Prime Target derivation	A8	Key assumptions
A4	12-month value bridge	A9	Experiment design

Figures are directional and based on the stated base-case assumptions.

Compass AI scored four actions for Minh Vy – including doing nothing.

Same trigger as slides 05–06. The decision is a comparison against silence, not a rule that fires on the skip limit.

THE USER IN FRONT OF THE ENGINE

THE CANDIDATE SET

UPLIFT FIGURES ARE ILLUSTRATIVE

MINH VY

FREE · DAILY LISTENER · PLAYLIST CURATOR

01 PERSUADABILITY

CANDIDATE FOR CAUSAL EVALUATION

One action may change the upgrade decision

02 TIMING · 5:55 PM

- High engagement — daily active listener
- Curating a playlist, repeated skips
- Skip limit reached — friction is live

03 RELEVANCE — INFERRED NEED

CONTROL, NOT DISCOVERY

04 SALIENCE — HOW IT IS SHOWN

In-app prompt naming unlimited skips, at the moment the limit is hit.

ALL GUARDRAILS CLEAR

Eligibility + contact-policy gates clear

GENERATE → EVALUATE every eligible action is built, then scored against silence			
CANDIDATE ACTION	NEED FIT	EXPECTED ACTION UPLIFT vs. SILENCE	VERDICT
Unlimited skips Standard Premium · in-app prompt	Control — match	+6.1 pp	SERVE
Ad-free listening Standard Premium · in-app prompt	Control — partial	+2.3 pp	Rejected — lower uplift
AI DJ discovery Premium trial	Discovery — no match	+1.4 pp	Rejected — wrong need
Do nothing Stay silent, protect the session	—	0.0 pp	The bar every action must clear

DECISION RULE

Serve only when the best eligible action creates more expected value than silence after costs and guardrails.

WHAT GETS LEARNED

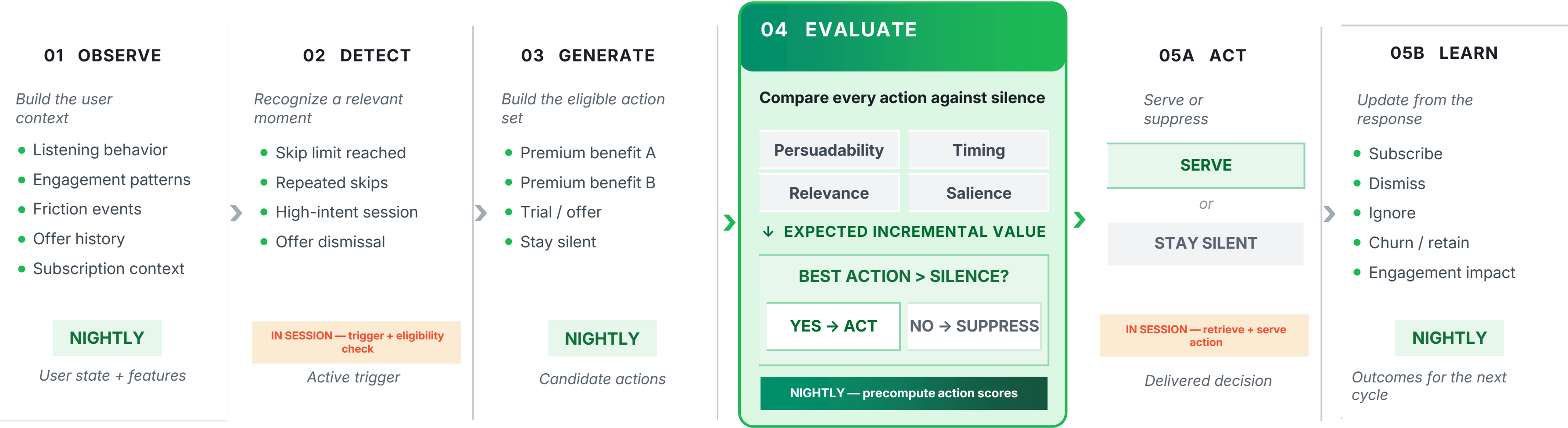
Subscribe, dismiss or ignore — read against the holdout, then folded into tonight’s rescoring.



KEY TAKEAWAY: Minh Vy gets unlimited skips because it beat three alternatives — including silence.

Compass AI converts behavioral signals into a governed next-best decision.

Most decisioning is prepared in batch; live sessions detect context, execute the precomputed action, and feed outcomes back into the next cycle.



 **FEEDBACK LOOP** Outcomes from LEARN refresh the user state that OBSERVE builds for the next decision cycle

GUARDRAILS Frequency cap · Quiet hours · Free-tier protection · Eligibility · Fairness · Brand safety

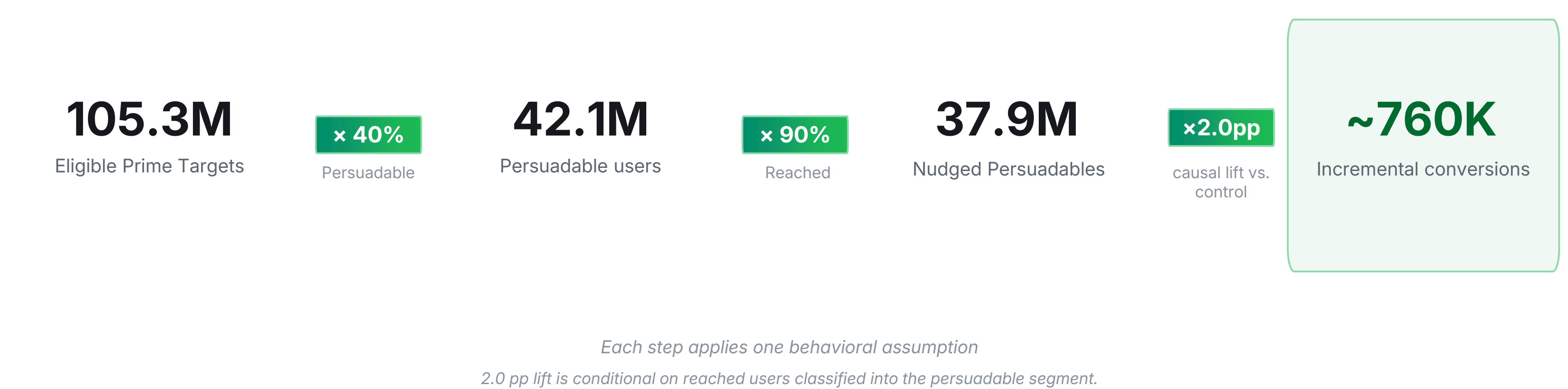
MEASUREMENT Persistent randomized control

BUILT ON EXISTING SPOTIFY CAPABILITIES Behavioral data · Feature infrastructure · Experimentation platform · Owned in-app surfaces · Model deployment

 **KEY TAKEAWAY:** Compass AI turns user context into a governed next-best action — and closes the loop with every response.

Three behavioral assumptions translate the eligible cohort into ~760K incremental conversions.

Only reached Persuadables generate modeled incremental conversions; Sure Things and non-responsive users add no value.

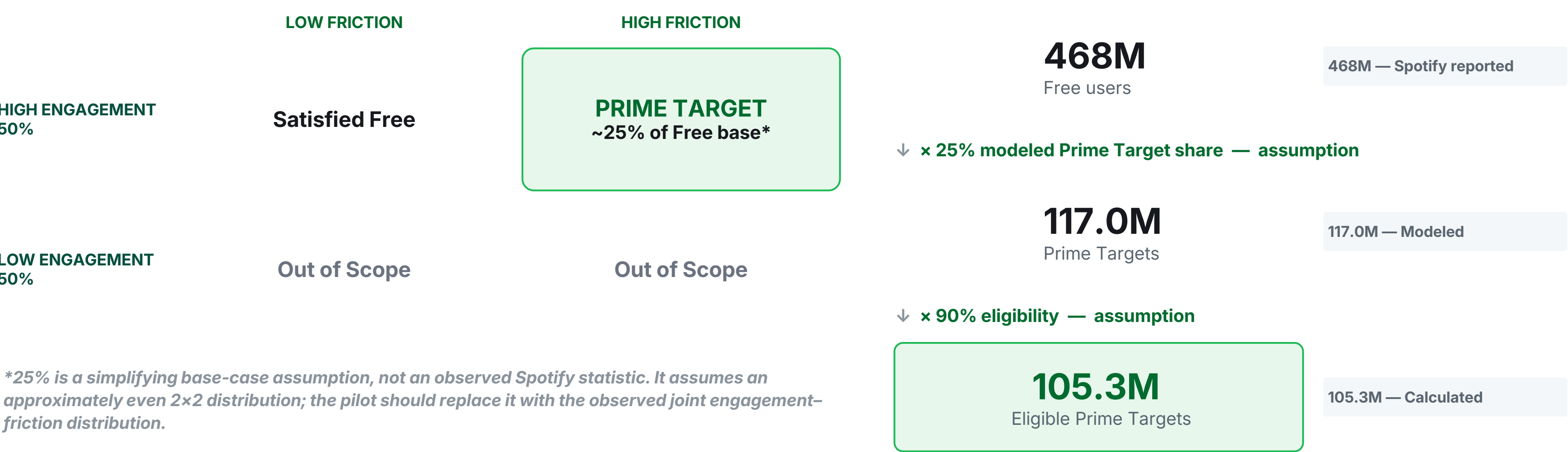


 **KEY TAKEAWAY:** Persuadability, reach, and incremental lift - not the size of Spotify's Free base alone- drive the conversion case.



A simple 2x2 screen models 25% of Free users as Prime Targets.

Median cutoffs on engagement and friction create four segments; the high-engagement / high-friction quadrant defines the modeled target pool.



~€14.7M of modeled 12-month gross-profit uplift translates into €4.7M of net benefit.

The model remains positive after fully loading both the €6.0M build and 12-month operating costs.

~760K incremental conversions × €19.46 12-month net GP per conversion	= €14.7M Gross-profit uplift
LESS	
Build investment (one-time)	– €6.00M
Base-case 12-month operating cost	– €4.00M
NET 12-MONTH BENEFIT	= €4.7M

FULLY LOADED 12-MONTH ROI
€4.7M net benefit
÷
€10.00M total 12-month cost
46.9%
12-month return on investment

€19.46 represents 12-month economics per incremental conversion.

Incremental subscribers generate strong unit economics even after fully charging the one-time build.

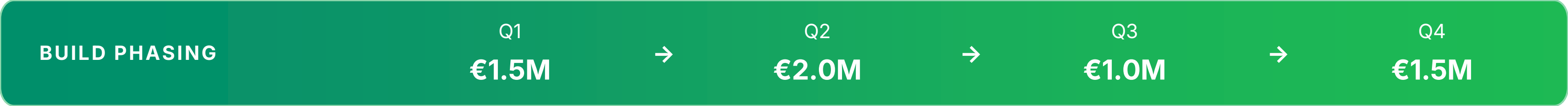
Lifetime gross profit is more than seven times the fully allocated acquisition cost.



The fully loaded case charges the entire €6.0M build; steady-state excludes that one-time cost.
Gross-profit LTV = monthly subscriber GP ÷ monthly churn; base churn ≈1.7%.

Recurring contribution repays the €6.0M build investment in ~6.7 months after scaled operation begins.

At-scale build recovery measures how long scaled operation takes to repay the upfront build after ongoing operating costs—not CAC payback per subscriber.



At-scale build recovery ≠ fully loaded CAC payback.

The former asks when the build is recovered once scaled operation begins; the latter allocates all costs to each conversion.



KEY TAKEAWAY: The upfront investment is recovered quickly under the base case, while staged funding limits capital at risk before proof.

One-way thresholds show headroom—but profitability depends on the assumptions jointly.

Each break-even reference holds the other drivers at base case; the economic decision evaluates persuadability, reach, and lift together.

ONE-WAY BREAK-EVEN REFERENCES

ALL OTHER ASSUMPTIONS HELD AT BASE CASE

INCREMENTAL LIFT

Base case 2.00pp



PERSUADABLE SHARE

Base case 40.0%



DELIVERY REACH

Base case 90.0%



FULL-INVESTMENT JOINT BREAK-EVEN
Required normalized product ≥ 0.664

$$(P / 40\%) \times (R / 90\%) \times (\text{Lift} / 2.0\text{pp})$$

BASE CASE

$$40\% \times 90\% \times 2.0\text{pp}$$

1.000 ✓ Clears hurdle

ILLUSTRATIVE GATE COMBINATION

$$33\% \times 80\% \times 1.33\text{pp}$$

0.488 ✗ Below hurdle

Operating guardrails and the one-way lift reference can all be met while full-investment economics still fail.

Each bar plots the base case on a fixed axis: light green = performance required to reach break-even, solid green = headroom above break-even, grey = unused axis. Q3 scale decision is forward-looking and evaluates remaining investment plus recurring costs; see slide 9.



KEY TAKEAWAY: Individual thresholds diagnose where performance is weak; only the joint economics determine whether the business case clears.

Six assumptions carry the business case - and each has a clear validation path.

The pilot is designed to replace modeled assumptions with observed evidence before scale capital is released.

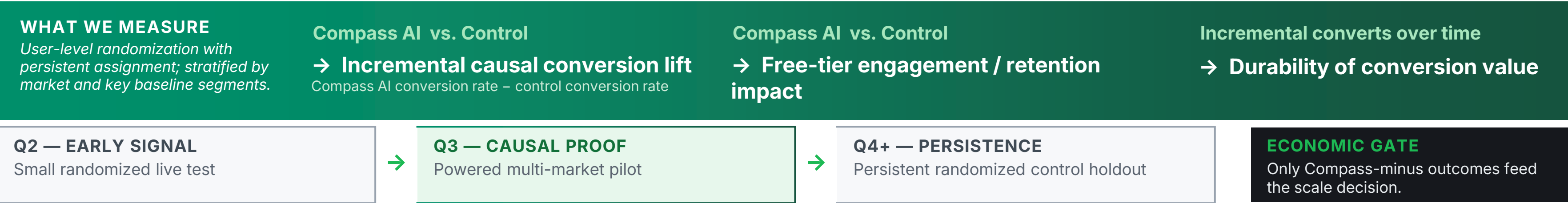
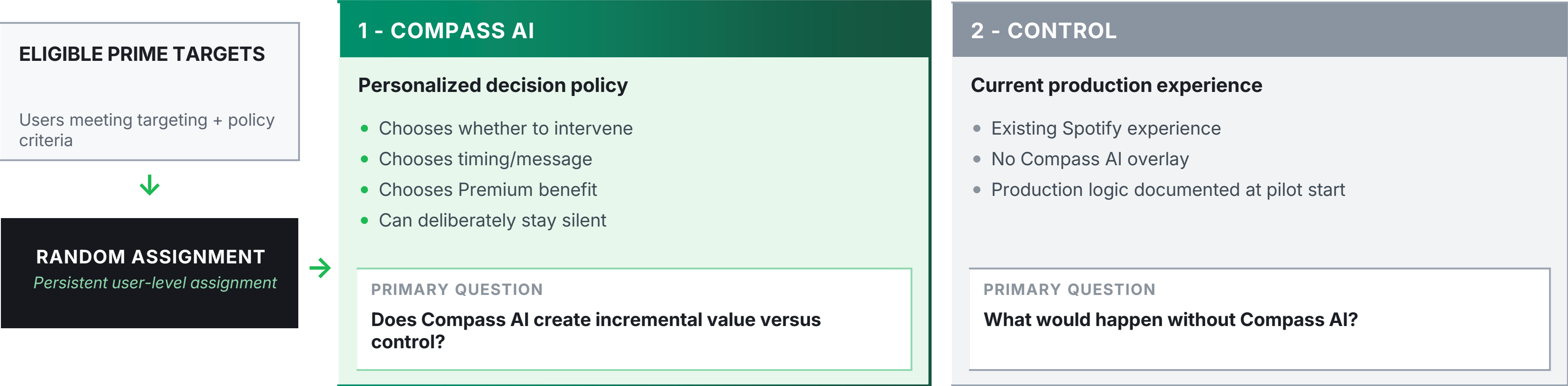
ASSUMPTION	BASE CASE	WHY IT MATTERS	HOW WE VALIDATE	VALIDATION SOURCE	WHAT MUST HOLD MODEL TODAY 40% × 90% × 2.0pp ↓ Q3 OBSERVED EVIDENCE ↓ SCALE / DO NOT SCALE Evidence replaces the model before capital scales.
Eligible Prime Target share	22.5% of the Free base	Sets the size of the addressable Free pool	Observed engagement × friction distribution + eligibility	PILOT VALIDATES	
Predicted positive-uplift share	40%	Determines how much of the eligible pool Compass AI can actually influence	Randomized Q3 pilot ≥33% operating gate	PILOT VALIDATES	
Delivery reach	90%	Determines how many Persuadables actually receive the intended intervention	Exposure / impression logs ≥80% operating gate	PILOT VALIDATES	
Incremental conversion lift vs. control	2.0pp	Primary driver of incremental conversions	Randomized Q2/Q3 experiment	PILOT VALIDATES	
12-month net GP / conversion	€19.46	Converts incremental subscriptions into GP uplift	Premium GP, displaced ad GP, retention	FINANCIAL VALIDATION	SCALE / DO NOT SCALE Evidence replaces the model before capital scales.
12-month operating cost	€4.00M	Recurring cost of running Compass AI at scale	Actual pilot infrastructure + offer cost	FINANCIAL VALIDATION	

WHAT CHANGES OUR MIND
Scale only if observed lift, persuadability, reach, and Free-tier impact jointly support positive fully loaded 12-month economics.
90% eligibility belongs to cohort derivation; 90% delivery reach is a separate conversion assumption. The pilot need not reproduce every base case exactly—it must clear the combined economic hurdle.

 **KEY TAKEAWAY:** Compass AI’s economics depend on a small number of measurable assumptions; the pilot is designed to turn each into observed evidence before scale.

A two-arm randomized test measures Compass AI against Spotify's actual production experience.

Control is defined and logged at pilot launch, so the test does not assume how Spotify promotes Premium today.





KEY TAKEAWAY: Compass AI earns credit only for outcomes above Spotify's actual production experience—not an assumed no-intervention baseline.